

PRADEEP DIGUVIDI NARAYANAPPA

📍 San Jose, CA | ✉ diguvidi.pradeep@gmail.com | 📞 +1 (669)-321-6111

🌐 linkedin.com/in/pradeep-diguvidi | 🐙 github.com/pradeepdn | 🌐 pradeepdn.github.io/pradeepdn-blog/

Summary

Software Engineer with **4+ years of experience** building **LLM-powered applications**, **AI agents**, **distributed systems**, and **enterprise software** across retail, healthcare, and nonprofit organizations. **Skilled** in translating **ambiguous customer requirements** into **production-ready systems** through **system design**, **AI engineering**, and **full-stack software development**. Experienced with **Retrieval-Augmented Generation (RAG)**, **LLM orchestration**, and **AI security**, delivering end-to-end solutions from technical discovery through production deployment with measurable business impact.

Professional Experience

CrowdDoing

AI Engineer

San Francisco

Mar. 2025 – Present

- **Developed** an **LLM-powered customer support agent** that automated enterprise FAQ workflows using **Retrieval-Augmented Generation (RAG)**, improving response quality while reducing manual response effort through context-aware answer generation for 1,000+ monthly support queries.
- **Engineered** an **agentic workflow** using **LangGraph**, **LangChain**, **MCP**, **Function Calling**, and **Tool Calling** to orchestrate multi-step reasoning, external tool execution, and adaptive customer interactions across **4 frontier LLMs**.
- **Integrated LiteLLM** as an **LLM gateway** with API gateways to orchestrate **GPT-4.1**, **GPT-4o**, **Claude**, and **Gemini**, supporting **multi-model routing**, **model fallback**, and **real-time web search**.
- **Monitored** LLM token consumption using LiteLLM analytics and configured per-user, team, and organization token limits to improve usage governance and cost visibility.
- **Deployed** AI services using **Docker**, **Kubernetes**, and **AWS**, integrating **Redis semantic caching** to reduce redundant LLM requests and support scalable inference workloads.
- **Strengthened** production security through **prompt injection detection**, input/output validation, **OAuth 2.0**, **JWT authentication**, **RBAC**, secure secret management, and API authorization, protecting customer-facing **AI applications** from unauthorized access and prompt manipulation.

TATA Consultancy Services (TCS)

Software Engineer

Bengaluru, India

Sep. 2021 – Jun. 2023

- **Led** technical discovery with retail stakeholders to deliver a full-stack commerce platform, integrating fragmented legacy systems and generating **\$2.5M** in seasonal revenue.
- **Redesigned** the MySQL data access layer and optimized distributed cloud services using indexing, query tuning, distributed caching, and cloud integration strategies, improving analytics response times by **32.6%** and reducing end-to-end latency.
- **Modernized** healthcare appointment scheduling by designing a scalable platform that maintained reliability during a **36%** increase in daily active users while strengthening multi-tenant security through data isolation and access controls that eliminated cross-tenant data leakage.
- **Automated** operational workflows using Bash-based validation checks and container orchestration alerts, saving **4.3 engineering hours per week** across recurring environment maintenance.

Systems Engineer

Aug. 2019 – Sep. 2021

- **Delivered** healthcare deployment automation and interactive clinical dashboards, improving release reliability, accelerating clinical operations by **43.5%**, and contributing to **\$3.5M** in project revenue.
- **Improved** relational data models and architected a Kafka-based streaming pipeline processing **1M+ daily healthcare records**, increasing core system performance by **41.2%** while enabling real-time hospital capacity monitoring without data loss.

Projects

Edu-Genius (Asynchronous LLM Orchestration Platform)

Jan. 2025 – Dec. 2025

- **Engineered** an **LLM-powered learning assistant** with **GPT-4o**, **streaming responses**, **persistent conversational context**, and **asynchronous orchestration**, supporting **200+ concurrent users** with **sub-400 ms latency** using **FastAPI**, **LangChain**, and **React**.
- **Built evaluation pipelines** measuring **latency**, **hallucination rates**, and **contextual consistency** across **500+ benchmark conversations**, achieving **94.2% contextual accuracy** for production-ready AI interactions.

Predictive Fleet Analytics Workspace

Aug. 2024 – Dec. 2024

- **Partnered** with transportation stakeholders to build a **predictive analytics platform** that transformed incomplete collision datasets into **actionable routing insights**, improving operational decision making with **96.83% model accuracy**.

Enterprise Security Monitoring Dashboard

Aug. 2024 – Dec. 2024

- **Developed** an **anomaly detection platform** using **Random Forest models** to help security teams identify **infrastructure risks** from high-volume system logs, enabling **operational visibility** through **machine learning-driven threat prioritization**.

Education

SAN JOSE STATE UNIVERSITY

Master of Science in Computer Engineering

San Jose, CA

Aug. 2023 – Dec. 2025

NEW HORIZON COLLEGE OF ENGINEERING

Bachelor of Engineering in Computer Science and Engineering

Bengaluru, India

Aug. 2015 – May. 2019

AI Certifications (Click on for credentials)

Databricks – Academy Accreditation – AI Agent Fundamentals 

Databricks – Academy Accreditation – Generative AI Fundamentals 

OpenAI – OpenAI Agents and Workflow Certification 

Anthropic – Agent Skills Certification 

Hugging Face – AI Agents Course Certification 

Technical Skills

AI Infrastructure: Docker, Kubernetes, AWS, Redis, OAuth 2.0, JWT, API Gateway, Prompt Injection Defense.

AI & Agentic Systems: LangGraph, LangChain, MCP, Function Calling, Tool Calling, Retrieval-Augmented Generation (RAG), LiteLLM, OpenAI Responses API, GPT-4.1, GPT-4o, Claude, Gemini, AI Agents.

Languages: Python, TypeScript, JavaScript, SQL, Java, C++

Systems: Distributed Systems, System Design, Microservices, REST APIs, Apache Kafka, Redis

Cloud & Infrastructure: AWS, Kubernetes, Docker, Jenkins CI/CD, Linux, Shell Scripting

Frameworks: FastAPI, React, Django, Angular, Flask, Streamlit

Data: PostgreSQL, MySQL, MongoDB, AWS DynamoDB, Google BigQuery, ETL/ELT, Pandas, NumPy, Scikit-learn

Awards: Innovation Excellence Award (TCS), Best Performance Award (TCS), Outstanding Achievement in Platform Efficiency (TCS)